OFFICIAL SYSTEM SPECIFICATION & OPERATING MANUAL

SmartAttendance™ Edge Biometric Terminal

Complete End-to-End Operating Guide: Hardware, Power, Enrollment, Attendance Scanning, Scheduling, Database Management, Charging & Telegram Cloud Synchronization

- **Edge AI Core:** Espressif ESP32-P4 Dual-Core RISC-V @ 400 MHz with 32 MB 200 MHz Hex-PSRAM
- **Display & Vision:** 7.0″ 1024×600 IPS Touchscreen (MIPI-DSI) and OmniVision MIPI-CSI Camera
- **Power & Charging:** USB-C 5V/3A, Onboard LiPo Charging Circuit, STC8 Gas Gauge, Auto-Protection
- **Zero-Cloud Biometrics:** 100% On-Device Face Detection, Landmark Alignment & MobileFaceNet (< 1s)
- **Robust Enrollment:** 30-Frame K-Means Pose Clustering, Adaptive Outlier Filter, Quality Scorecard
- **Lecture Scheduling:** Conversational Telegram Bot Wizard (`/schedule`) & REST Cloud Sync API
- **Dual Telegram Ecosystem:** Student Check-in Push Alerts and Lecturer Session Summaries
- **Database & Reporting:** Offline SQLite3 (`/sdcard/attendance.db`) with On-Device CSV Export

Document Version:	2.0.0 (Comprehensive Edition)	**Hardware Platform:**	CrowPanel Advanced 7.0″ ESP32-P4 v1.0
Firmware Base:	ESP-IDF v5.3+ / ESP-DL v1.1.4+	**Display Controller:**	EK79007 (MIPI-DSI) + GT911 Touch
Target Audience:	Administrators, Lecturers, IT & Facility Staff	**Security Standard:**	Offline Edge Biometric Vector Storage
Publication Date:	October 2026	**Document ID:**	SA-ESP32P4-UM-2026-FULL

[v] BEST PRACTICE: Document Purpose & Compliance

This user manual provides the authoritative operational, electrical, and administrative instructions for the entire SmartAttendance system ecosystem. It covers physical installation, power/charging management, biometric enrollment, classroom attendance scanning, lecture scheduling via Telegram, database maintenance, and disaster recovery.

Table of Contents

Chapter 1: System Overview & Architecture	Product philosophy, Edge AI capabilities, system block diagram
Chapter 2: Hardware Architecture & Pin Interfaces	ESP32-P4 SoC, display bridge, touch, camera, audio, pin mappings
Chapter 3: Power Architecture, Battery & Charging	USB-C, battery management, charging indicators, power-saving, shutdown
Chapter 4: Commissioning & Initial Setup	Unboxing, MicroSD card preparation, Setup Wizard, Master Admin PIN
Chapter 5: Biometric Face Recognition Engine	ESP-DL inference, 5-point alignment, 128D embeddings, dynamic margin
Chapter 6: Facial Enrollment Procedures	On-device UI, SoftAP Web Portal, 8-step quality evaluation scorecard
Chapter 7: Lecture Scheduling & Course Management	Telegram Bot `/schedule` wizard, FastAPI Cloud sync, active session
Chapter 8: Daily Attendance Scanning Operations	Main scanner UI, verification badges, multi-frame voting, debounce logic
Chapter 9: User Database & Roster Administration	Viewing users, search, role editing, non-destructive re-enrollment, delete
Chapter 10: Attendance Records, Reports & CSV Export	Session filtering, on-screen reports, SD card CSV export, cloud push
Chapter 11: Telegram Bot Cloud Ecosystem	Student check-in alerts, lecturer portal, admin remote control commands
Chapter 12: System Settings, Maintenance & OTA Updates	Backlight PWM, audio volume, Wi-Fi configuration, dual-bank OTA
Chapter 13: Troubleshooting & Fault Diagnostics	Hardware recovery, camera init, touch quirks, SD mount, error codes
Chapter 14: Technical Specifications & Safety Ratings	Electrical ratings, temperature limits, mechanical mounting, care

System Architecture Overview

The **SmartAttendance** ecosystem operates on a decoupled edge-cloud architecture designed specifically for educational institutions and corporate enterprises. The physical hardware terminal performs 100% of face detection, alignment, feature extraction, and database matching on local RISC-V silicon. The cloud service operates as an asynchronous coordination layer (Telegram Bot + FastAPI sync engine) that distributes lecture schedules to terminals and broadcasts instant push notifications to students upon successful verification.

1. PHYSICAL TERMINAL	2. EDGE AI PIPELINE	3. PERSISTENCE ENGINE	4. TELEGRAM & CLOUD
• 7″ IPS Touch (MIPI-DSI) • SC2336 Camera (MIPI-CSI) • LiPo Battery & STC8 Gauge • MicroSD FAT32 Storage • USB-C 5V/3A Power	• ESP-DL HumanFaceDetect • 5-Point Affine Warping • MobileFaceNet int8 Model • 128D Cosine Similarity • Dynamic Margin Gate	• SQLite3 Embedded DB • /sdcard/attendance.db • CSV Attendance Logs • User Biometric Vectors • NVS Configuration	• Student Push Alerts • Lecturer /schedule • Admin Remote Console • Web AP register.html • Dual-Bank OTA Updates

Chapter 1: System Overview & Technical Specifications

The SmartAttendance terminal is built on the **Elecrow CrowPanel Advanced 7.0″ ESP32-P4 v1.0** development platform. The hardware combines high-speed MIPI interfaces with dual-core processing power to execute complex deep learning inference on edge silicon without external accelerators.

Component / Subsystem	Technical Specification & Architecture Details
Processor (SoC)	Espressif ESP32-P4 dual-core RISC-V @ up to 400 MHz (LP Core @ 40 MHz)
System Memory (PSRAM)	32 MB Hex-Mode (16-line) PSRAM clocked at 200 MHz; external data bus
Internal Flash	16 MB Quad-SPI Flash (6 MB factory app partition, NVS, PHY calibration)
Display Panel	7.0″ IPS TFT LCD, 1024×600 native resolution, EK79007 MIPI-DSI bridge (2 data lanes @ 900 Mbps, DPI clock 51 MHz)
Touch Controller	Goodix GT911 Capacitive Touch (I2C address 0x5D, SDA: GPIO45, SCL: GPIO46, RST: GPIO40, INT: GPIO42)
Backlight Control	Hardware PWM on **GPIO31** (30 kHz frequency, 11-bit resolution, formula: duty = percent*18 + 200)
Camera Subsystem	OmniVision SC2336 (24-pin MIPI-CSI, SCCB 0x36), 1280×720 RAW8 @ 30fps, hardware ISP Bayer BGGR
MicroSD Card Interface	SDMMC 1-bit mode (CLK: GPIO43, CMD: GPIO44, D0: GPIO39), FAT32 mounted at /sdcard
External RTC (Optional)	DS1302 Real-Time Clock on J7 Header (CLK: GPIO51, DAT: GPIO50, CE: GPIO49) with CR2032 coin backup
Connectivity	Wi-Fi 802.11 b/g/n (2.4 GHz) via companion wireless bridge; SoftAP mode for captive enrollment portal
Power Architecture	5V DC via USB Type-C or screw terminals; onboard LiPo battery charger with STC8 telemetry gauge

Chapter 2: Hardware Architecture & Pin Interfaces

Proper electrical connections and pin configurations are vital during assembly and maintenance. The table below details critical hardware interfaces:

Interface	GPIO Pin	Signal Type	Functional Notes & Electrical Behavior
Backlight PWM	GPIO 31	PWM Output	High-frequency 30 kHz dimmer; 0% disables backlight, 100% full illumination.
Touch I2C SDA	GPIO 45	Open Drain	External 4.7kΩ pull-up required; legacy I2C driver on I2C_NUM_0.
Touch I2C SCL	GPIO 46	Open Drain	External 4.7kΩ pull-up required; clocked at 100 kHz standard speed.
Touch Reset (RST)	GPIO 40	Output Push-Pull	Driven manually during boot to lock GT911 address at 0x5D.
Touch Interrupt (INT)	GPIO 42	Output Push-Pull	Low level during reset pulse; releases high to complete address locking.
SD Card Clock	GPIO 43	Clock Output	SDMMC 1-bit high-speed clock line.
SD Card Command	GPIO 44	Bidirectional	SDMMC command/response line with internal pull-up.
SD Card Data 0	GPIO 39	Bidirectional	SDMMC data line D0 for 1-bit sector transfers.
Battery ADC Sense	GPIO 4 (ADC_CH4)	Analog Input	Voltage divider monitoring terminal battery voltage (3.3V to 4.2V).

Chapter 3: Power Architecture, Battery Management & Charging

The SmartAttendance terminal is engineered for dual-mode power operation: stationary wall-powered continuous operation and portable, battery-powered deployment for mobile classrooms, field examinations, and seminar venues.

3.1 Power Supply Inputs & Requirements

- **Primary USB Type-C Input:** 5.0 V DC ±5%, 3.0 A recommended. Powers the terminal and simultaneously charges the internal battery.
- **Auxiliary Screw Terminal (VIN / GND):** 5.0 V regulated DC input for fixed wall-mounted kiosk installations using enterprise DIN-rail power supplies.
- **Peak Consumption Profile:** During boot, camera burst capture, and Wi-Fi sync, current spikes reach **2.2 A**. A substandard 500mA PC USB port will induce brownout resets.

3.2 Lithium-Ion Battery & STC8 Gas Gauge Architecture

The CrowPanel Advanced ESP32-P4 incorporates an onboard Lithium-Polymer (LiPo) / Lithium-Ion charging management IC paired with an STC8 micro-controller acting as an intelligent fuel gauge:

Parameter	Operating Value / Threshold	System Behavior & Enforcement
Battery Chemistry	Li-Po / Li-Ion (1S / 3.7V nominal)	Compatible with standard single-cell 3.7V packs (2500mAh to 10000mAh capacity).
Full Voltage (100%)	4.20 V DC	Charging terminates automatically. Battery icon shows full solid green bar.
Empty Voltage (0%)	3.30 V DC	Baseline cutoff voltage. Device initiates protected emergency shutdown.
Low Battery Warning	15% State of Charge	Battery icon turns flashing amber/red. An advisory banner prompts: 'Battery Low (15%). Please connect charger.'
Critical Shutdown	5% State of Charge	System pauses camera tasks, flushes SQLite transactions to MicroSD, safely unmounts FAT32, and powers down to prevent database corruption.
Telemetry Polling	10-Second Interval	ADC channel 4 and STC8 coprocessor read cell voltage and compute smoothed moving average.

3.3 On-Screen Battery & Charging Indicators

The top-right status bar of the LVGL touchscreen provides continuous real-time power telemetry:

- **Charging Active (Cyan Lightning Bolt):** When a 5V USB-C charger is connected, the battery symbol switches to LV_SYMBOL_CHARGE in glowing cyan (#00CFFF), accompanied by the live percentage (e.g., [bolt] 74%).
- **Normal Discharging ( ):**
  - **90% – 100%:** LV_SYMBOL_BATTERY_FULL (4 bars)
  - **60% – 89%:** LV_SYMBOL_BATTERY_3 (3 bars)
  - **30% – 59%:** LV_SYMBOL_BATTERY_2 (2 bars)
  - **10% – 29%:** LV_SYMBOL_BATTERY_1 (1 bar)
- **Critical Low (Flashing Red Empty):** Below 10%, the icon displays LV_SYMBOL_BATTERY_EMPTY in bold red (#FF4444).

3.4 Recommended Charging & Battery Maintenance Procedures

- **Standard Full Charge Cycle:** Connect a 5V/3A USB-C charger. A fully discharged 5000mAh battery requires approximately **2.5 to 3.5 hours** to reach 100% full capacity.
- **Charging While Operating:** The device supports pass-through charging. It can conduct classroom attendance scanning while connected to power without overheating.
- **Storage Recommendations:** If storing the terminal inactive between academic semesters, charge the battery to approximately **50% – 60%** and store in a cool, dry room (15°C – 25°C). Recharge every 3 months to prevent cell degradation.
- **Battery Calibration Logging:** For technical audits, firmware includes a calibration logger (`BATTERY_CALIB_LOGGING=1` in `config.h`) that logs terminal mV and STC8 readings to `/sdcard/logs/batt_calib.csv` every 20 seconds.

[!] IMPORTANT: Battery Safety Rules

Do NOT puncture, expose to open flame, or crush the internal battery pack. If the battery shows physical swelling, immediately disconnect the USB-C charger, power down the device, and replace the cell with a manufacturer-approved 3.7V LiPo pack equipped with an integrated over-voltage/under-voltage protection circuit (PCM).

Chapter 4: Commissioning & Initial Setup

Follow these procedures to prepare the MicroSD card, execute the First Boot Setup Wizard, and establish system administration security.

4.1 MicroSD Card Preparation & Directory Layout

The terminal requires a high-speed MicroSD card (UHS-I Class 10, FAT32 format). Before powering on the device, create the following directory structure on the card:

```
/sdcard/
  |-- models/p4/
  |     |-- human_face_detect_msr_s8_v1.espdl   (Face detection stage 1)
  |     |-- human_face_detect_mnp_s8_v1.espdl   (Face landmark alignment stage 2)
  |     `-- mfn_s8_v1.espdl                     (MobileFaceNet 128D embedding model)
  |-- attendance.db                            (SQLite database; auto-created on boot)
  |-- attendance_backup.db                     (Periodic database snapshot)
  |-- reports/                                 (Directory for exported attendance CSV files)
  `-- logs/                                    (System telemetry & battery calibration logs)
```

4.2 First Boot & Interactive Setup Wizard

When powered on with a fresh database, the terminal automatically launches the **Interactive Setup Wizard**:

- **Step 1: Hardware Self-Test:** The firmware validates 32MB Hex-PSRAM, MIPI-DSI display bridge, GT911 touch coordinates, camera sensor, and mounts `/sdcard`. All subsystems must show a green checkmark.
- **Step 2: Master Administrator PIN:** Set a secure **4-digit Master PIN** (default initial PIN is 1234). This PIN protects User Management, Attendance Reports, and System Settings.
- **Step 3: Institution Identity:** Enter the Department Name (e.g., Computer Science) and Terminal Location (e.g., Lecture Theatre 1). These are saved to NVS and appear on report headers.
- **Step 4: Date, Time & RTC Sync:** Configure your timezone offset (e.g., UTC+1). When connected to Wi-Fi, time is synchronized via SNTP. If an external DS1302 RTC battery is installed, time is preserved across reboots.
- **Step 5: Wi-Fi Provisioning:** Select your institution's 2.4 GHz Wi-Fi network and enter the passphrase using the on-screen keyboard, or tap **[Skip Network]** for 100% offline deployment.

Chapter 5: Biometric Face Recognition Engine

SmartAttendance executes real-time machine vision entirely on the ESP32-P4 RISC-V cores. Understanding the biometric pipeline ensures optimal lighting and positioning.

5.1 Real-Time Machine Vision Pipeline

Pipeline Stage	Algorithm / Engine	Operation & Quality Criteria
1. Acquisition	MIPI-CSI Camera	Streams 1280×720 RAW8 frames at 30 fps. Double-buffered in SPIRAM (`s_detection_fb_buf`) to prevent DMA tearing.
2. Face Detection	ESP-DL HumanFaceDetect	Quantized int8 neural network detects faces in 320×240 preview. Bounding box must be ≥ 30px with confidence ≥ 0.60.
3. Landmark Alignment	5-Point Keypoint Warping	Extracts left eye, right eye, nose, left mouth, right mouth. Warps facial crop to a canonical 112×112 RGB image.
4. Quality Filtering	Laplacian, Yaw & Glare Gates	Discards blurry frames (sharpness < 5.0), yaw > 35°, specular glare (>15% pixels Y>240), and open mouth ratio (>0.85).
5. Embedding Extraction	MobileFaceNet int8	Compresses 112×112 aligned face into a compact, normalized 128-dimensional signed integer vector embedding.
6. Multi-Frame Consensus	Dynamic Margin Voting	Accumulates 6 frames (or early-exit 3 unanimous frames ≥ 0.80). Validates tiered margin over runner-up before confirming identity.

5.2 Tiered Dynamic Margin Anti-Collision Gate

To prevent identity confusion between candidates with similar facial features (e.g., siblings or similar bone structures), the system enforces a **Dynamic Separation Margin**:

- **High Similarity (S ≥ 0.78):** High-confidence match. Requires a separation margin of $\Delta S \geq 0.03$ over candidate #2.
- **Mid Similarity (0.74 ≤ S < 0.78):** Standard match. Requires a separation margin of $\Delta S \geq 0.05$ over candidate #2.
- **Borderline Similarity (0.70 ≤ S < 0.74):** Borderline match. Requires a strict separation margin of $\Delta S \geq 0.08$ over candidate #2. If margin is insufficient, identity is withheld as ambiguous.

Chapter 6: Facial Enrollment Procedures

Facial enrollment registers an individual's biometric profile into the local SQLite database. SmartAttendance features an 8-step statistically robust template synthesis algorithm that avoids single-frame capture errors.

6.1 Enrollment Method 1: Direct On-Device Touch UI

- **Open User Manager:** From the main screen, tap the top menu icon, select **User Management**, and enter the 4-digit Master Admin PIN.
- **Input User Information:** Tap **[+ Add User]**. Enter Full Name, Matric / Staff ID (e.g., CSC/2022/1045), Phone Number (international format: +2348012345678), and assign Role (Student, Lecturer, or Admin).
- **Position Candidate:** Guide the user to stand 0.5 to 0.8 meters directly in front of the camera, looking at the display with a neutral facial expression.
- **30-Frame Capture:** Tap **[Start Face Capture]**. The device captures 30 quality-verified facial frames while the progress bar advances from 0% to 100%.
- **Quality Scorecard:** Upon completion, the screen displays the enrollment quality score (e.g., 89/100 EXCELLENT) and writes the synthesized vector to SQLite.

6.2 Enrollment Method 2: SoftAP Captive Web Portal (`register.html`)

For bulk classroom registration without crowding the terminal touch screen, administrators can use any smartphone or laptop:

- **Connect Wi-Fi:** On your mobile device, connect to the terminal's Wi-Fi hotspot: Attendance_Setup.
- **Access Web Portal:** Open a web browser and navigate to http://192.168.4.1/register.html.
- **Enter Session PIN:** Enter the dynamic 4-digit session PIN shown on the physical terminal screen banner.
- **Queue Candidate:** Fill in the student's name, matriculation number, department, and phone number, then tap **[Queue for Capture]**.
- **Automatic Camera Trigger:** The student on-site looks at the terminal screen. The device detects the queued candidate, captures the 30-frame sequence, displays a success prompt, and confirms back to the admin's phone.

6.3 The 8-Step Statistically Robust Enrollment Algorithm

Step	Algorithmic Function	Mathematical & Embedded Implementation
Step 1	Sample Acquisition	Collects N=30 valid frames. Rejects frames if face count≠1, sharpness<5.0, yaw>35°, or mouth ratio>0.85.
Step 2	PSRAM Storage	Stores all 30 extracted 128D vectors in PSRAM-backed buffer array, eliminating SRAM stack overflow.
Step 3	Adaptive Outlier Removal	Calculates 30×30 cosine similarity matrix. Rejects samples where mean similarity $m_i < \max(\mu - 1.5\sigma, 0.65)$.
Step 4 & 5	K-Means Pose Clustering	Runs K=3 K-Means clustering on inliers to group frontal, slight left, and slight right poses. Prunes empty clusters.
Step 6	Cluster Weighted Fusion	Synthesizes centroid vector weighted by cluster size. Rescales vector to mean L2 target norm and quantizes to int8.
Step 7	Quality Scoring (Q)	Computes quality index: Q = 0.40*R_inlier + 0.40*C_compact + 0.20*S_sharp (0 to 100 scale).
Step 8	Persistence & Feedback	Stores the single representative vector in SQLite. Displays score badge; enables 'Redo' button ONLY if POOR (<55).

[v] BEST PRACTICE: Duplicate Face Prevention

During enrollment, the terminal automatically compares the newly synthesized vector against the entire enrolled user cache. If cosine similarity matches an existing student ($S \geq 0.70$), enrollment halts and displays: 'Face already registered! Cannot create duplicate identity.'

Chapter 7: Lecture Scheduling & Course Session Management

SmartAttendance integrates an automated lecture scheduling system. Lecturers create and manage class timetables directly through the official Telegram bot, which asynchronously synchronizes with the offline physical terminal.

7.1 Lecturer Verification & Authentication

Before scheduling lectures, instructors verify their identity via Telegram:

- Open Telegram and search for the institutional bot: @MyUniAttendance_Bot.
- Send the /start command.
- Tap the **[Share Contact Details]** button. The bot normalizes the instructor's phone number and matches it against the lecturer profiles synchronized from the physical terminal.
- Upon successful verification, the instructor's Telegram ID is linked to their profile, unlocking scheduling permissions.

7.2 The 5-Step Scheduling Wizard (`/schedule`)

To schedule a lecture session, the instructor sends /schedule to the bot. The conversational wizard guides them through five simple steps:

Step	Prompt Field	Example Input	Validation & Formatting Rules
Step 1	Course Code	CSC 301	Alphanumeric department course code (e.g., CSC301, EEE402).
Step 2	Course Title	Database Systems	Full descriptive name of the lecture course.
Step 3	Session Date	24/10/2026	Format: DD/MM/YYYY. Cannot be set to past dates.
Step 4	Start Time	09:00	Format: HH:MM in 24-hour clock (e.g., 09:00, 14:30).
Step 5	End Time	11:00	Format: HH:MM. Must be chronologically after Start Time.

7.3 Device Schedule Synchronization (`GET /api/get_schedules`)

- **Cloud Storage:** The FastAPI backend stores active schedules in bot_data.db.
- **Device Pull:** When the terminal connects to Wi-Fi (either automatically every 6 hours or when an admin taps **[Sync Now]**), it queries: GET /api/get_schedules?since=<timestamp>.
- **Automatic Session Activation:** When the terminal's internal clock matches a scheduled lecture window, the main screen automatically updates its header banner to the active course code (e.g., ACTIVE SESSION: CSC 301). All attendance scans during that window are tagged with that course session ID.
- **Schedule Deletions:** If an instructor cancels a class via Telegram, the deletion is logged in schedule_deletions and pruned from the terminal during the next sync.

> **[i] NOTE: Ad-Hoc (Unscheduled) Classes**
> If a lecture occurs outside scheduled timetables, the lecturer or administrator can tap **[Select Course]** directly on the terminal touch screen and pick any registered course from the drop-down list to open an immediate manual attendance session.

Chapter 8: Daily Attendance Scanning Operations

During class sessions, the terminal operates autonomously in **Attendance Scanning Mode**. Students walk up to the device, look at the screen, and receive immediate visual confirmation.

8.1 Screen Layout & Visual Feedback

The 7.0″ IPS display is divided into three functional zones:

- **Top Telemetry Bar:** Displays the institutional logo, current time (synchronized via SNTP/RTC), active course code (e.g., CSC 301: Database Systems), Wi-Fi connection indicator, and battery status with live percentage.
- **Center Live Camera Viewfinder:** Streams high-definition 30 fps video. Detected faces are tracked dynamically with a color-coded bounding reticle:
  - **GREEN RETICLE:** Verified match! Displays student name, matric ID, and match confidence.
  - **AMBER RETICLE:** Evaluating multi-frame consensus; user is prompted to hold still.
  - **RED RETICLE:** Unknown face or quality violation (excessive glare, wide open mouth, or motion blur).
- **Bottom Status Banner:** Displays total attendance counter for the active class (e.g., Verified: 54 / 62 Students) and real-time guidance prompts.

8.2 Sub-Second Verification & Debounce Logic

• **Rapid Verification:** Complete face detection, landmark affine warp, MobileFaceNet inference, and database query execute in **under 1.0 second**.
• **Early Exit Path:** If three consecutive frames yield unanimous match scores of $S \geq 0.80$, the system triggers an immediate early-exit confirmation without waiting for the full 6-frame window.
• **15-Minute Duplicate Debounce:** Once a student is marked present, subsequent scans by the same student within 15 minutes display: "Already Recorded: [Student Name]" without creating duplicate database entries.

8.3 Environmental Guidance & Anti-Spoofing Filters

Quality Filter	Threshold Trigger	Visual Prompt & System Action
Specular Glare Gate	Luminance Y > 240 on >15% pixels	Prompts: "Glare detected — adjust angle away from light". Prevents flash/sunlight identity distortion.
Mouth Opening / Jaw Gate	Mouth height / Eye dist > 0.85	Prompts: "Please close mouth & look straight". Rejects laughing, talking, and yawning distortions.
Motion Sharpness Gate	Laplacian variance < 5.0	Prompts: "Blurry image — please hold still". Prevents blurred edge matching.
Extreme Head Pose	Yaw angle $\|\theta\| > 35.0°$	Prompts: "Please face the camera directly". Rejects profile angles.

Chapter 9: User Database & Roster Administration

Authorized administrators manage user profiles via the on-device **User Management Module**:

- **Roster Browser:** Displays a searchable card list of all registered students, lecturers, and administrators. Each card summarizes Full Name, Matric / Staff ID, Role tag, Registration Date, and Enrollment Quality Score.
- **Safe Re-Enrollment Protocol:** If a student changes their appearance (e.g., new glasses, beard, or facial modification) and recognition confidence drops, select their profile and tap **[Re-enroll Face]**. Firmware v1.1.4+ executes a **non-destructive in-place update**: it captures a new 30-frame template and updates the facial vector in SQLite without modifying their user ID, historical attendance records, or Telegram linkages.
- **User Deletion & Archive:** Tapping **[Delete User]** requires Master PIN confirmation. Deleting a user removes their biometric template to free memory while preserving past attendance logs under their original matric ID for historical audit compliance.

Chapter 10: Attendance Records, Reports & CSV Export

The terminal maintains immutable audit trails stored locally in SQLite (`/sdcard/attendance.db`) and exports structured CSV reports for university administrative systems.

10.1 On-Device Reports Browser

Open **Attendance Reports** from the main menu (requires Master PIN). Administrators and lecturers can filter records by **Date** or **Course Code**. The interface presents total enrolled count, verified present count, absent count, and class attendance percentage.

10.2 Exporting CSV Reports to MicroSD Card

Tap the **[Export CSV to SD]** button. The terminal generates a standard UTF-8 Comma-Separated Values file at /sdcard/reports/attendance_YYYYMMDD.csv with the following schema:

```
Session_ID,Date,Time,Course_Code,Course_Title,Student_ID,Full_Name,Role,Confidence,Status
1042,2026-10-21,09:02:14,CSC301,Database Systems,CSC/2022/1045,Adesare Adegbagi,Student,0.842,PRESENT
1042,2026-10-21,09:03:01,CSC301,Database Systems,CSC/2022/1088,Solomon Israel,Student,0.811,PRESENT
1042,2026-10-21,09:04:22,CSC301,Database Systems,CSC/2022/1102,Fatima Bello,Student,0.875,PRESENT
1042,2026-10-21,09:05:40,CSC301,Database Systems,CSC/2022/1150,Chukwuma Obi,Student,0.829,PRESENT
```

10.3 Cloud Push Synchronization (`POST /api/sync_attendance`)

When the device connects to Wi-Fi, it pushes all un-synced attendance events to the institutional backend via POST /api/sync_attendance. Once acknowledged, the local SQLite database marks the records as synchronized (synced = 1).

Chapter 11: Dual Telegram Bot Ecosystem

The Telegram integration provides a zero-maintenance cloud bridge serving students, lecturers, and system administrators.

- **Student Push Notifications:** When a student is verified at the terminal, the cloud bot instantly dispatches a personal Telegram notification: *'Hello Adesare! Your attendance has been recorded for CSC 301 (Database Systems) at 09:02 AM in Lab Block A.'*
- **Lecturer Real-Time Class Summaries:** Upon class conclusion, the bot compiles an automated summary message for the lecturer, detailing attendance percentage and attaching the complete CSV roll sheet.
- **Administrative Remote Control Commands:** Restricted strictly to authorized administrator Telegram chat IDs:

Command	Parameters	Description & Operational Response
/status	None	Returns terminal uptime, CPU temperature, free PSRAM/heap, active Wi-Fi SSID, and enrolled user count.
/sync	None	Commands terminal to connect to Wi-Fi, upload pending attendance records, and pull new schedules.
/export	[course] [date]	Generates and directly uploads the attendance CSV document into the admin Telegram chat.
/roster	[course_code]	Lists enrolled students for the specified course and displays semester attendance percentages.
/ota	[firmware_url]	Triggers Over-The-Air firmware upgrade; terminal downloads, validates SHA256, flashes, and reboots.
/reboot	None	Executes a graceful software restart of the SmartAttendance hardware terminal.

Chapter 12: System Settings, Maintenance & OTA Updates

The **System Settings** screen (Master PIN required) provides access to hardware tuning and lifecycle maintenance:

- **Display Backlight & Dimming:** Adjust screen brightness from 10% to 100% via the verified 30 kHz PWM driver on GPIO31. Enable **Auto-Dimming** (1 to 15 minutes) to conserve battery during unattended periods.
- **Wi-Fi Network Configuration:** Scans for 2.4 GHz access points, displays live RSSI signal strength meters, and saves credentials in encrypted ESP-IDF NVS storage.
- **Dual-Bank Over-The-Air (OTA) Updates:** The terminal features two 6 MB application partitions (ota_0 and ota_1). Firmware updates can be applied via the Local Web Portal (http://192.168.4.1/update) or remotely via the Telegram /ota command. If an update fails CRC verification, the bootloader automatically rolls back to the stable firmware bank.
- **Database Backup & Disaster Recovery:** The terminal automatically maintains a backup at /sdcard/attendance_backup.db. If hardware suffers catastrophic physical damage, removing the MicroSD card and inserting it into a replacement CrowPanel ESP32-P4 board restores 100% of user profiles and history instantly.

Chapter 13: Troubleshooting & Fault Diagnostics

Consult this diagnostic troubleshooting matrix compiled from hands-on embedded test engineering:

Observed Fault / Symptom	Probable Root Cause	Actionable Resolution Procedure
Screen dark; no power LED	Inadequate power source or tripped battery PCM.	Connect a certified 5V 3A USB-C power supply. Do not power from 500mA PC USB ports. Power LED must illuminate blue.
Touchscreen unresponsive	GT911 I2C address conflict or missing RST/INT pulse.	Power cycle terminal. Ensure no conductive metal touches display bezel. Re-flash firmware with verified GT911 0x5D init sequence.
Camera shows black preview or init fails	24-pin MIPI-CSI ribbon unseated or wrong SCCB address.	Power down device, unlatch 24-pin FPC connector, confirm gold pins face PCB, firmly reseat cable, and snap connector down.
'SD Card Mount Failed' banner	Card unformatted, corrupted FAT32, or missing models.	Reformat MicroSD on PC as FAT32 (32KB clusters). Confirm /models/p4/ contains all three .espdl neural models.
Frequent 'Unknown Face' rejections	Severe backlight glare, dim lighting, or poor template.	Reposition device away from direct sunlight. Check student score in User Manager; if rating is POOR (<55), perform a re-enrollment.
Rapid battery drain (< 2 hours)	Backlight at 100% constant or degraded battery cell.	Set backlight to 60% and enable 2-minute auto-dim timeout. If issue persists, replace cell with a genuine 3.7V LiPo pack.
Telegram notifications not delivered	Wi-Fi offline or phone number not linked in bot.	Check Wi-Fi icon on status bar. Have student verify contact card in @MyUniAttendance_Bot to link their Telegram ID.

Chapter 14: Technical Specifications & Safety Warnings

Electrical & Environmental Parameter	Official Specification / Operating Boundary
Input Operating Voltage	5.0 V DC (± 5%) via USB Type-C or screw terminal block
Operating Current Consumption	Nominal: 1.1 A to 1.6 A; Peak Burst: 2.2 A
Operating Ambient Temperature	0°C to +50°C (32°F to 122°F) non-condensing
Storage Temperature	-20°C to +70°C (-4°F to 158°F)
Relative Humidity Limits	10% to 90% non-condensing
Display Viewing Angle	178° Horizontal / 178° Vertical full IPS viewing angle
Verification Distance Range	Optimal: 0.5 m to 0.9 m; Usable: 0.3 m to 1.4 m
Biometric False Acceptance Rate (FAR)	< 0.001% (at default similarity threshold 0.70 with dynamic margin)
Biometric False Rejection Rate (FRR)	< 1.5% (under recommended indoor ambient lighting 150 - 500 lux)

14.2 Physical Installation & Maintenance Precautions

- **Mounting Height:** For standard standing operation, mount terminal centerline at **1.45 to 1.55 meters** from the floor.
- **Cleaning:** Clean the 7.0″ glass capacitive touch panel with a soft microfiber cloth lightly moistened with optical screen cleaner. Never apply alcohol, solvents, or abrasive powders.
- **Thermal Clearance:** Ensure the rear of the CrowPanel enclosure has at least 15mm convective air clearance.

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